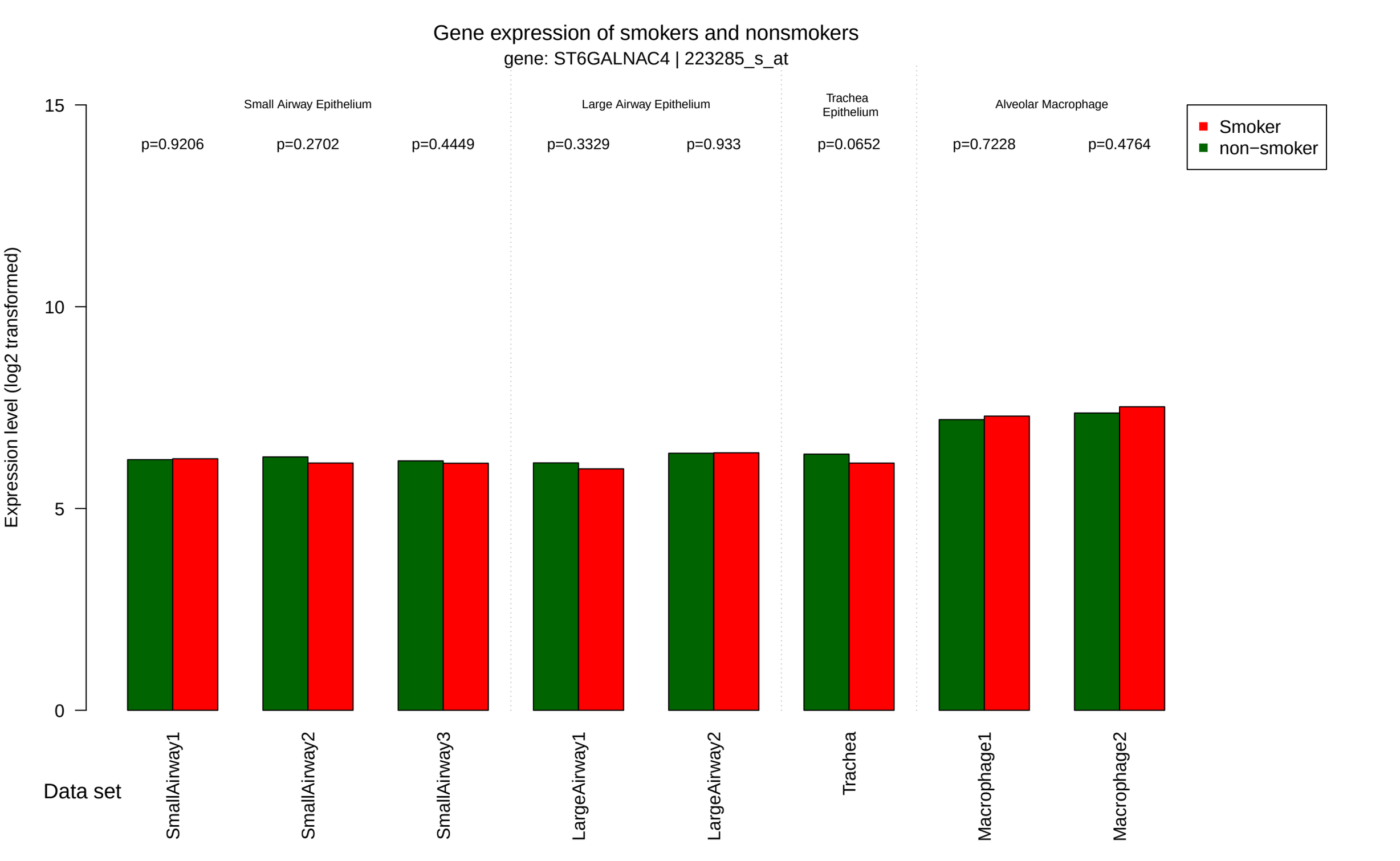
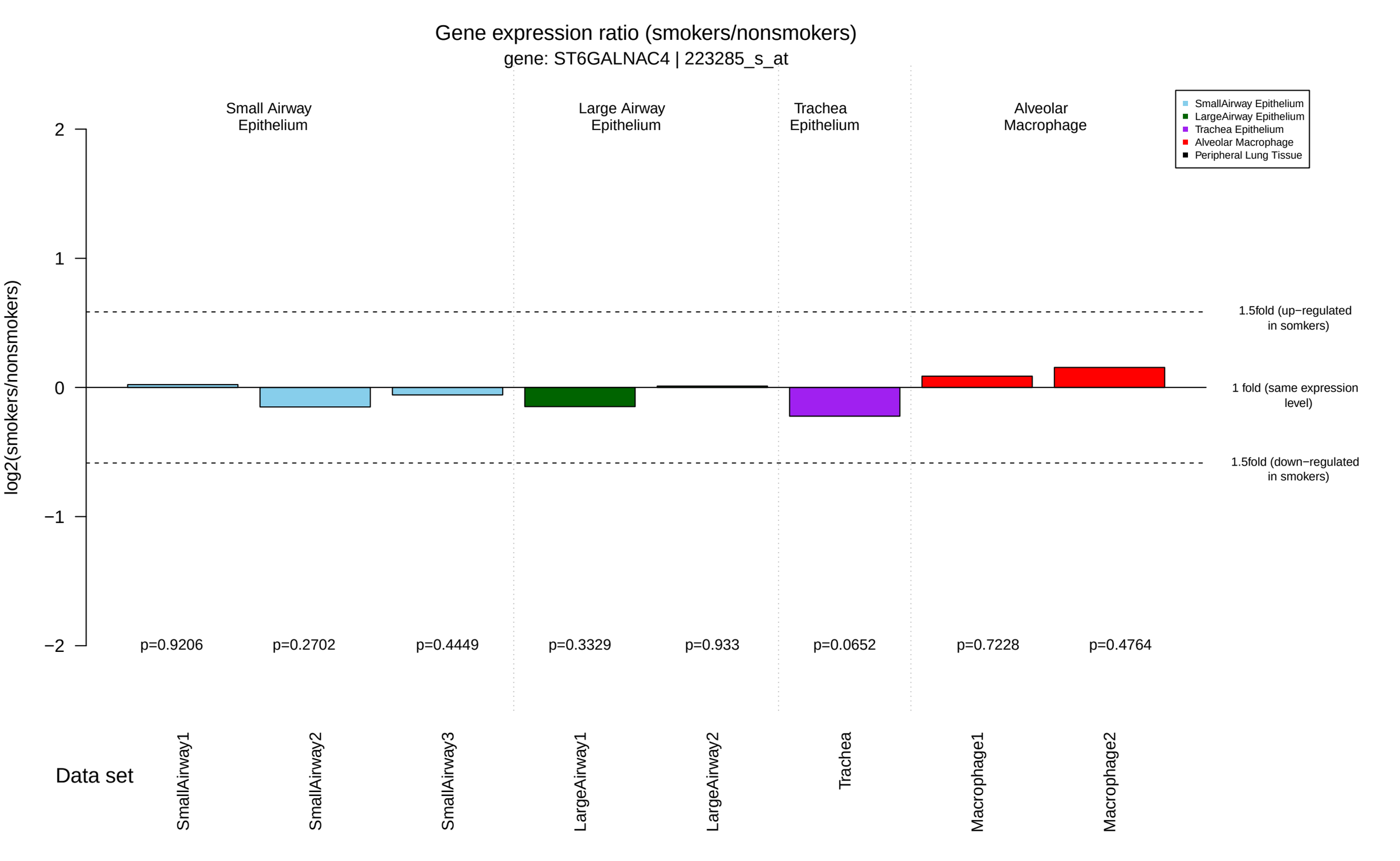
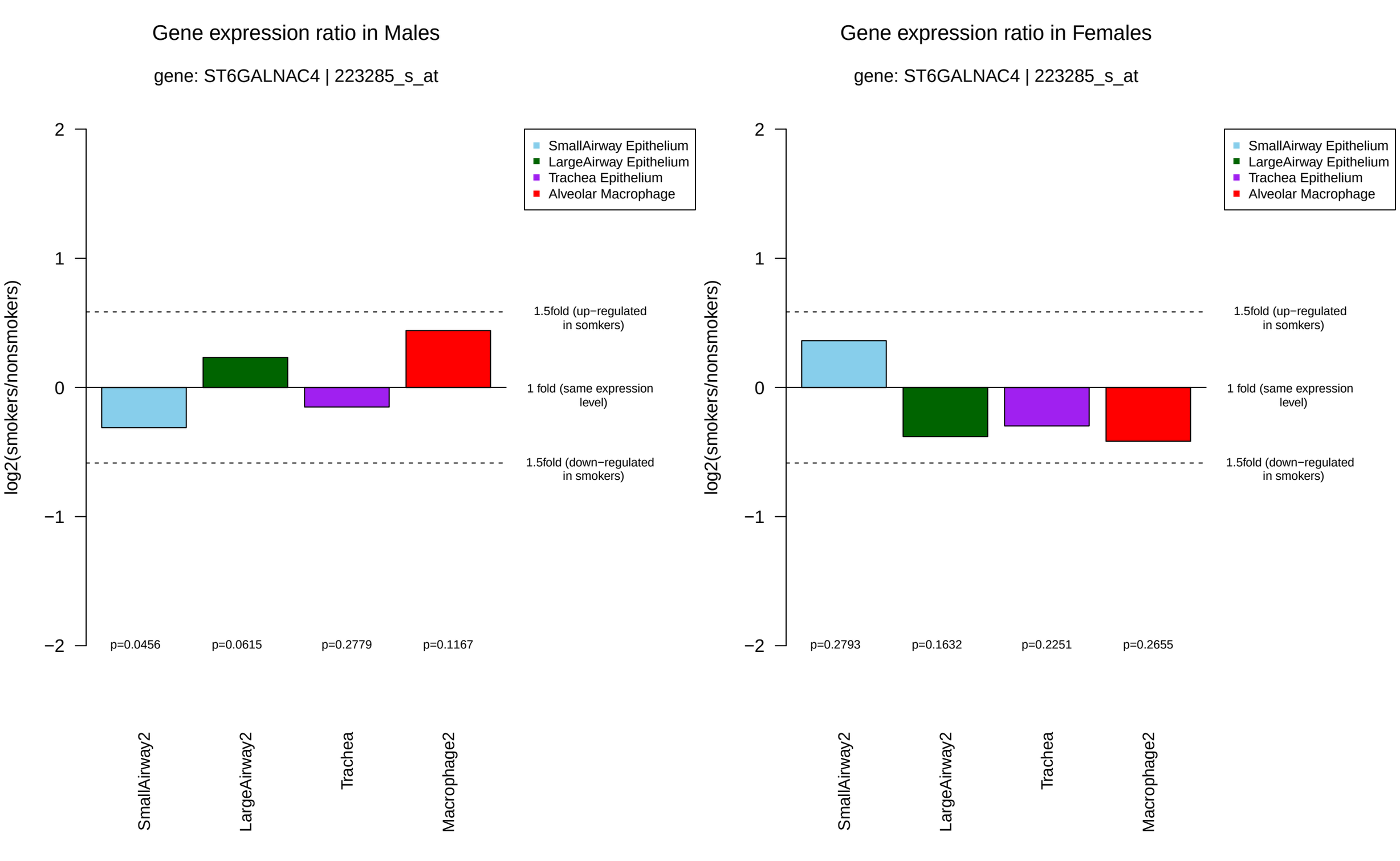
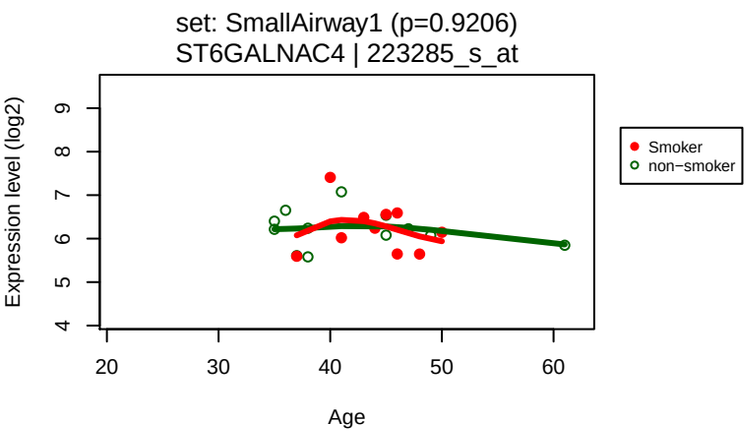
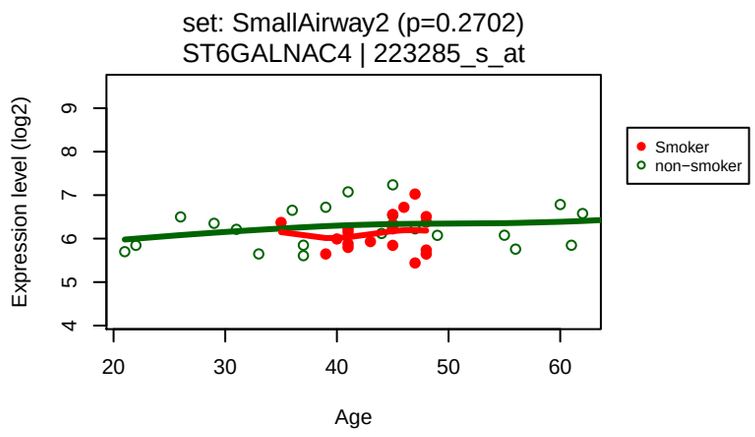
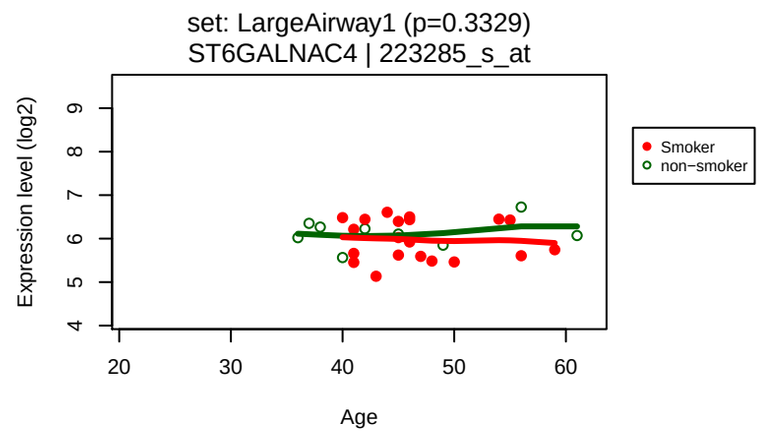

Gene expression of smokers and nonsmokers
gene: ST6GALNAC4 | 223285_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9206
p=0.2702
p=0.4449
p=0.3329
p=0.933
p=0.0652
p=0.7228
p=0.4764
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

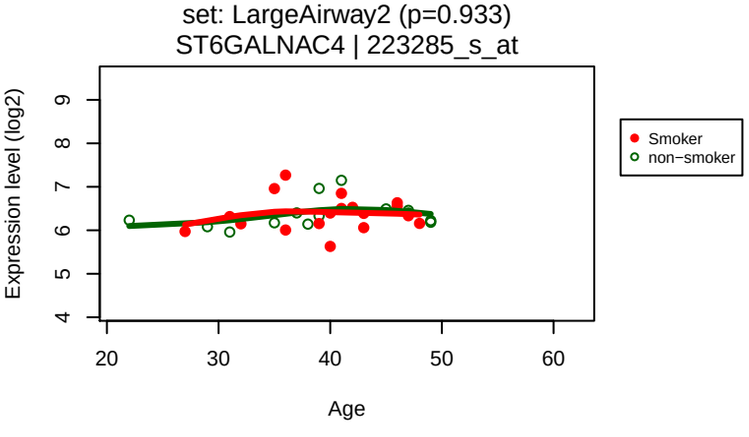

Gene expression ratio (smokers/nonsmokers)
gene: ST6GALNAC4 | 223285_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9206
p=0.2702
p=0.4449
p=0.3329
p=0.933
p=0.0652
p=0.7228
p=0.4764
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

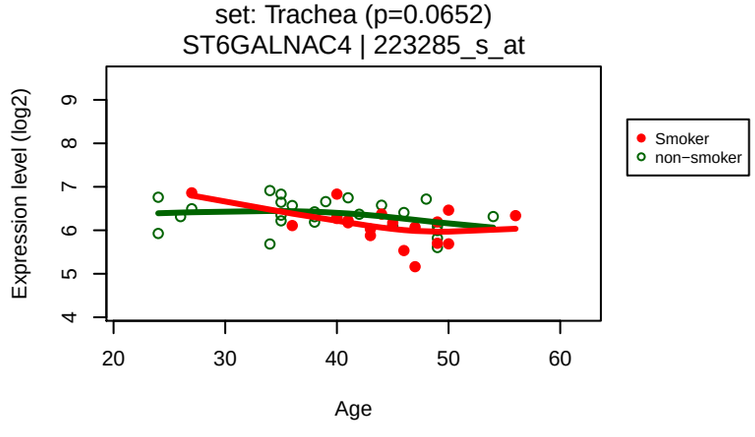

Gene expression ratio in Males
gene: ST6GALNAC4 | 223285_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0456
p=0.0615
p=0.2779
p=0.1167
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST6GALNAC4 | 223285_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2793
p=0.1632
p=0.2251
p=0.2655
SmallAirway2
LargeAirway2
Trachea
Macrophage2

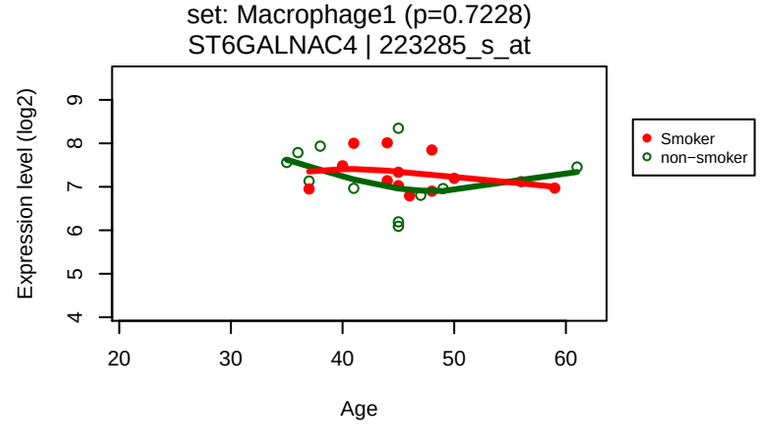

set: SmallAirway1 (p=0.9206)
ST6GALNAC4 | 223285_s_at
Expression level (log2)
Age
Smoker
non−smoker

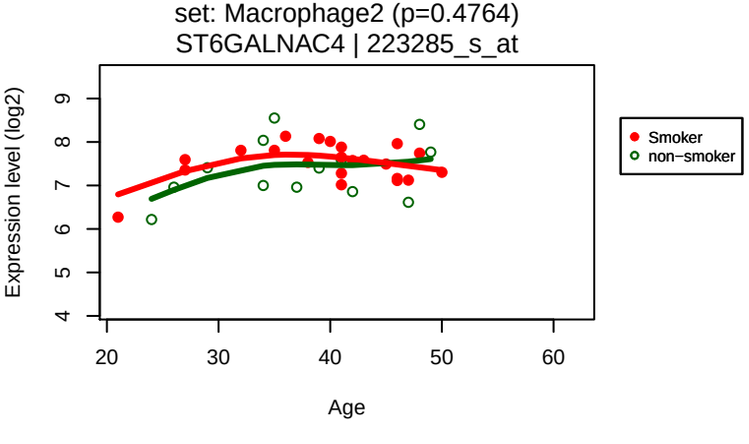

set: SmallAirway2 (p=0.2702)
ST6GALNAC4 | 223285_s_at
Expression level (log2)
Age
Smoker
non−smoker

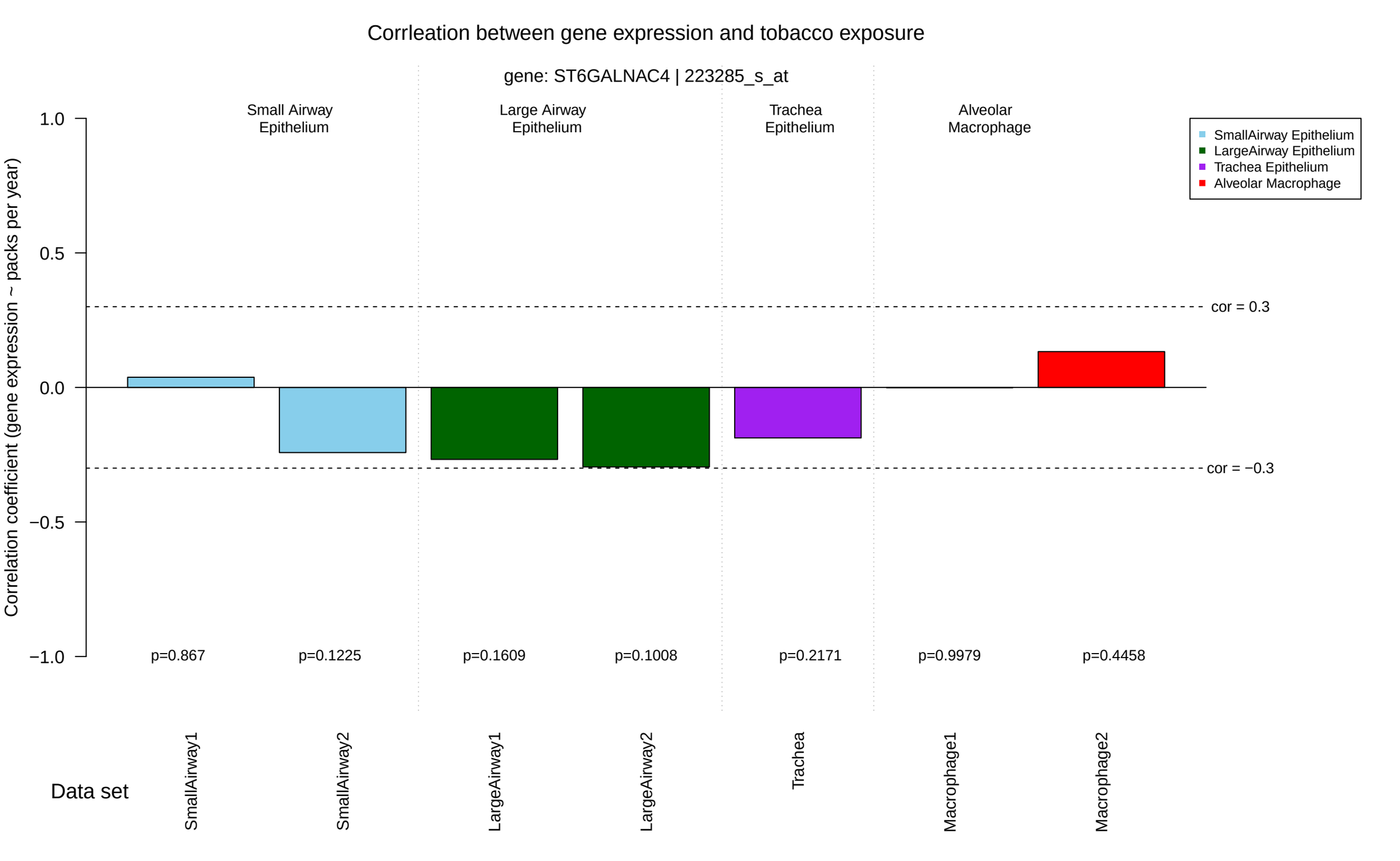

set: LargeAirway1 (p=0.3329)
ST6GALNAC4 | 223285_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.933)
ST6GALNAC4 | 223285_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0652)
ST6GALNAC4 | 223285_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7228)
ST6GALNAC4 | 223285_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4764)
ST6GALNAC4 | 223285_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST6GALNAC4 | 223285_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.867
p=0.1225
p=0.1609
p=0.1008
p=0.2171
p=0.9979
p=0.4458
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2